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Full Length Article

Protein interaction prediction for Alzheimer's disease using a multi-source protein features fusion framework

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ABSTRACT

Background: Studying the protein interactions associated with Alzheimer's disease can provide insights into the pathogenesis of this confounding disease. However, to date, researchers have only considered laboratory data and the scientific literature in isolation, and so have not been able to construct a comprehensive protein-protein interaction (PPI) network from which to predict potential interactions.
Methods: In this study, we devised a framework that integrates protein attributes and interaction information extracted from both experimental data and the scientific literature. Based on these data from multiple sources, We then constructed a PPI network reflecting a diverse range of node features and edge weights. Further, the Graph Convolutional Network (GCN) applicable to the network was used for the link prediction task.
Findings: Our proposed method achieved superior performance in protein interaction prediction for Alzheimer's disease with an AUC value of 0.8935. We identified the top ten results predicted by our model as the most promising potential protein interactions for Alzheimer's disease.
Interpretation: The analysis of the predictive results using empirical evaluation verify that our framework has produced a reasonable roadmap for discovering potentially novel protein interactions related to Alzheimer's disease.

1. Introduction

Proteins play a key role in many biological processes and cellular functions. However, these protein functions are rarely performed in isolation.[1] Rather, proteins typically fulfilll their purpose by interacting with other proteins in their surroundings.[2] Currently, common wisdom suggests that studying protein interactions may help elucidate the complex mechanisms and intricate pathways underlying neurological disorders such as Alzheimer's disease.

Alzheimer's disease is a progressive chronic neurological disorder that primarily affects the brain. It leads to a high incidence of disability and gradual loss of cognitive and behavioral functions.[3] The causes of Alzheimer's disease are multifactorial, but the exact pathogenesis of Alzheimer's disease remains elusive. The hope is that by studying protein interactions, we will find the etiology of the disease and then develop an effective therapeutic treatment.[4]

As we move into the postgenomic era, advances in high-throughput experimental methods allow for a precise quantitative description of protein interactions.[5] Given the intertwined nature of protein-protein interactions (PPIs), using computational approaches to analyze and data mine PPIs stands out as an effective method for studying protein functions.[6] Current PPI network-based prediction methods either use different protein features,[7] such as sequence features,[8] structure features,[9] function features[10] and comparative genomics features,[11] or machine learning and deep learning techniques.[12] For example, Taghipour et al.[13] mapped multiple physical protein interactions associated with Escherichia coli that were measured in the laboratory into a single weighted PPI network. From this, they predicted new protein complexes by detecting related functional proteins. Du et al.[14] extracted multiple protein features including: the fixed-length features associated with the composition, transition and distribution of amino acids; quasi sequence order descriptors; and amphiphilic pseudo amino acid composition (PAAC). They then generated predictions of novel protein interactions via a deep neural network called DeepPPI. However, these methods have some limitations. One is the length of time it takes to do laboratory research and the delay in publishing novel findings. Another is the

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narrow focus and singular dimension of the research. As such, we find very few multidisciplinary connections in this field. Importantly, adopting such a limited research perspective may influence the computation algorithms used to find protein interactions resulting in knowledge bias. The last limitation is the enormous cost of managing and storing the data.

With these shortcomings in mind, the recent explosive increases in the volume of biomedical literature are opening up new opportunities to solve these issues. Literature-based knowledge discovery could provide a widely accessible and promising pathway for exploring the genetic basis for diseases. Techniques such as co-occurrence analysis, meta-analysis, centrality measurement, and text mining have been widely used to synthesize and distill genetic knowledge from the literature. For example, Lee et al.[15] proposed a machine learning approach COMMODAR, which cooperatively used three discriminative features from multiple modalities to extract context-specific molecular relations from biomedical literature. Heo et al.[16] proposed a hybrid ranking method that combines entity co-occurrence with word embedding techniques to find hidden relationships pertinent to Alzheimer's disease. The literature represents a condensed review of previous studies and is supported by ample evidence, which could be used as highly credible and authoritative sources of information. Further, such studies tend to reflect the latest progress and findings in the field, making them highly innovative and cutting-edge. Additionally, they typically offer plentiful semantic information and cover various aspects and multiple features of this field. For these reasons, the literature can be a valuable source of data for discovering novel knowledge. Such advantages will facilitate more accurate and detailed knowledge discovery.[17]

Link prediction in protein interaction networks can be used to estimate the probability of interaction between protiens based on known information in the network. Through the prediction of interaction between proteins, costs will be drastically reduced while enhancing the research efficiency.[18]

There are three primary modes of link prediction: those that are similarity-based; those that are model-based; and embedding techniques. Similarity-based methods rank all non-connected links according to how likely they are to actually be connected by assigning a score to them. The score is generally based either on some structural feature, such as the number of common neighbors between two nodes, or on some other property, such as the network's community structure. There are also several different structural similarity measures, including those based on local information and those based on global information. Some of the well-known structure-based prediction methods include shortest distance, common neighbors, Jaccard's Index, Katz, and so on.[19] The model-based methods operate on the premise that the probability of a link existing between any node pairs relies on a certain set of parameters. As such, these methods seek to identify the optimal set of parameters for assessing the conditional probability of a link existing between two nodes. Typically, the correct set of parameters can be found from the network topology. Relational Bayesian networks and relational Markov networks are two examples that fall into this category.[20] More recently, the embedding techniques coupled with deep graph models have been the most popular approach to link prediction. This is because embedding approaches automate the process of feature engineering. More specifically, a network is mapped into a low-dimensional feature space such that similar nodes are placed near to each other. Here, the network components are represented by embedding vectors made up of real numbers. The next step is to create an embedding of the graph and, here, there are three main approaches. The first is matrix factorization, in which the graph is represented in the form of a matrix. The matrix is then factorized, and the embedding vectors for the nodes are obtained. This approach captures the information about the structure of the network structure and enables prediction by decomposing the network's adjacency matrix or the associated matrix into a low-dimensional potential factor matrix. The NMF-LP (non-negative matrix factorization link prediction) model is an instance of such a method.[21] Another approach is to learn the graph embedding via a deep learning method. Most of these methods use an autoencoder to reduce the dimensionality of the graph's structure.In this approach, embedding vectors that reflect node similarity are generated using graph structure information and node attribute information to achieve link prediction. For example, DNGR uses deep autoencoder to capture the non-linearity in a graph while simultaneously applying dimension reduction when constructing the graph embedding.[22] The last approach to learning the graph's embedding is based on random walks. A random walk can preserve most of a graph's structural properties, like node centrality and similarity. Therefore, they can also be used to learn an embedding for a network. Such methods learn low-dimensional embedded representations of nodes by modelling path traversal in the network to enable link prediction. Deepwalk and Node2Vec are the two most well-known approaches based on random walks.[23]

This study proposes a framework for predicting novel protein interactions associated with Alzheimer's disease. The framework (1) integrates multi-featured PPI data from experimental and literature sources. By leveraging the large amount of credible, professional and standardized knowledge of protein interactions in the literature, it reduces the impact of problems such as single source of PPI data, research lag and data noise, and provides high-quality and accurate data for knowledge discovery. (2) constructs a weighted and multi-dimensional PPI network. By incorporating rich node attributes and edge weights into the network, it helps the GCN model to learn the multi-dimensional features of proteins and the topology of PPI network in a more in-depth way, thereby improving the protein interaction prediction performance. (3) predicts novel protein interactions associated with Alzheimer's disease using an enhanced GCN model. The model outperforms single-source PPI prediction methods and state-of-the-art PPI prediction models, achieving an AUC value of 0.8935. Additionally, the model identified ten promising unknown protein interactions for Alzheimer 's disease, which demonstrates the effectiveness of the proposed method.

2. Data and methods

2.1. Data

Protein-protein interactions (PPIs) describe molecular contacts between proteins mediated by electrostatic forces or biochemical processes. These interactions exhibit remarkable diversity across different protein pairs, requiring multifaceted analytical approaches. Current characterization methods include comparative genomics, coevolution analysis, and structural modeling. In this study, we constructed a dataset of protein–protein interactions by integrating data from five distinct sources: basic semantic information extracted from the scientific literature, along with four categories of experimental data—biochemical properties, functional annotations, structural features, and gene co-expression profiles. To extract semantic information related to Alzheimer's disease, we searched the title and abstract fields in PubMed using "Alzheimer" as a MeSH term. As of 14 August 2023, this search yielded 119,682 documents. From these, we identified 6028 protein entities and extracted 194,168 protein–protein interactions. Because our analysis focused exclusively on human proteins, we cross-referenced the extracted entities with human protein data from the UniProt database. After filtering, we retained 5237 human proteins related to Alzheimer's disease. To construct the multi-feature protein interaction dataset from experimental data, we computed the interactions among these 5237 proteins. This resulted in a total of 1141,112 experimentally derived protein–protein interactions.

Protein attributes—such as physicochemical properties, structural organization, and functional roles—define the unique characteristics of each protein. We extracted five aspects of protein attributes for further study. We extracted amino acid sequences, functional descriptions, and gene ontology (GO) information from the Uniprot database. We extracted information about the protein families from the Interpro database. And we extracted information about protein expression levels from THE HUMAN PROTEIN ATLAS database. It is important to note that some of these attributes were derived from experimental data—specifically, amino acid sequences, protein families, and expression levels. In contrast, functional descriptions and GO annotations were curated from the scientific literature. Based on these combined data sources, we constructed a weighted, multi-feature protein–protein interaction (PPI) network comprising 5237 protein entities and 1225,616 interactions. This network served as the foundation for our subsequent link prediction analysis.

2.2. Protein interaction calculation

To compute protein–protein interactions, we primarily utilized four characteristic feature types: biochemical properties, structural features, functional annotations, and gene co-expression profiles. In addition to experimental data, we mined protein interaction information from the scientific literature to supplement laboratory-derived interactions. We also incorporated interaction weights to enrich the PPI network. This section explains the specifics of how we calculated the interactions from each data source.

2.2.1. Multi-feature protein interaction calculation based on the experiment data

We employed the MIscore metric to quantify protein–protein interactions derived from biochemical and experimental data. This score is calculated from the weighted sum of three different aspects, including the reports supporting the interaction, experimental detection methods and interaction types. Full details of how these scores are calculated are outlined in.[24] These calculations resulted in 981, 187 protein interactions.

In terms of the physical structure of the protein interactions, we applied a method grounded in thermodynamic principles—assuming that an energetically favorable interaction is more likely to be biologically valid. The calculation details were shown in,[25] and this analysis yielded 216,738 protein interactions.

To evaluate functional similarity, we assessed the semantic similarity of Gene Ontology (GO) terms associated with proteins participating in the same biological processes. As described in,[26] we adopted an improved hybrid algorithm that computes semantic similarity based on the topology of the GO directed acyclic graph (DAG). This resulted in 31, 735 protein interactions.

For the gene co-expression-based interactions, we used the co-expression data from each GEO series (GSE) sample on the Affymatrix platform as the main source. A Bayesian statistical framework was used to calculate the likelihood of a functional association between co-expressed gene pairs. Then the log-likelihood scores (LLS) of the co-expression were considered as correlation scores of the gene co-expression. See[27] for more details. These calculations resulted in 207, 297 protein interactions.

2.2.2. Protein interaction calculation based on the literature semantic

In this study, the semantic information of protein interaction was extracted from the scientific literature using a co-occurrence- based text mining technique. We assumed that if two proteins appear within the same sentence, the two proteins are likely to interact with each other. The MetaMap tool that is one of the primary indexing systems for the National Library of Medicine (NLM), was utilized to extract protein entities from literature. Notably, since some proteins have been found to have different expression, their names need to be standardized. Hence, we relied on unique protein numbers to standardize protein entities with the help of the Uniprot database. Then we sorted through the human proteins among them. After duplicate deleting and statistics conducting, a total of 6028 protein entities and 194,168 protein interactions were obtained. In addition to co-occurrence-based protein interaction extraction methods, there are also some rule-based, machine learning and deep learning extraction methods, but these methods suffer from the shortcomings of requiring a large amount of training data and more complex feature selection. The co-occurrence-based extraction method is simpler and can cover more comprehensive protein interaction information. However, considering that the co-occurrence-based method introduces noise, we incorporated semantic information into the data to further improve the accuracy of the protein interaction expressions. There are many types of protein interactions in the literature and different categories of protein interactions have distinct representations. Therefore, values could be assigned to protein pairs containing explicit protein interaction predicates between the corresponding two proteins in the same sentence to improve the semantic information they express. The predicate categorization and assignment used in this study was only for reference. In this study, we summarized the existing related works and classified the predicates into two categories.[28] The first category was any reporting of chemical changes linked to the interaction partners of the PPI("modifying interactions",MI), including methylation and demethylation and similarly phosphorylation and dephosphorylation as well as other types of chemical changes. These modifications can be subsumed as post translational modifications (PTMs), which are a subcategory of PPIs. These predicates are based on strong experimental proof, with explicit statements of the interaction patterns in the scientific literatureand therefore can be a clear indication of the interaction between two proteins. The second category denoted interactions without chemical changes ("non-modifying interactions", NMI). This group contains results such as one protein interacts, activates or binds another protein. These predicates are usually related to the reconstruct regulatory and signaling pathways of PPI. The classification of the two categories is motivated by the linguistic representation strength of predicates. The predicates in MIs have stronger scientific evidence to support the interactionindicating a greater ability to characterize protein interactions. Thus, they should be given a higher score. NMIs category contains some common predicates for PPI identification. Although the predicates of this category can't achieve the explicit mention of protein interaction pattern compared to the first category, they can accurately characterize the meaning of the PPI. Hence, we assigned a slightly lower score to these predicates compared to the first category. The predicates collected in the table are a comprehensive summary of previous relevant research. These predicates tend to appear more frequently in scientific publications and can accurately express the meaning of PPI. In total, we summarized 65 predicates - 24 in the first category with a score of 2, and 41 in the second category with a score of 1.75. The specific predicates in each category are shown in Table 1. The formula for calculating the literature semantic protein interaction score is as follows:

$$score = p_1 \times i + p_2 \times j + (p - p_1 - p_2) \times 1 \quad (1)$$

Table 1
A summary of predicates expressing the meaning of protein interaction under two categories of MI and NMI.

Category	Predicates for the expression of protein interactions
Modifying interactions (MI)	acetylat*, acylat*, amidat*, biotinylat*, brominat*, carboxylat*, cysteinylat*, dephosphorylat*, demethylat*, farnesylat*, formylate, hydroxilat*, hydroxylat*, methylat*, myristylat*, nitrosylat*, myristoylat*, palmitoylat*, palmitylat*, phosphorylat*, pyruvat*, sumoylat*, ubiquitinat*, ubiquitinylat*
Non-modifying interactions (NMI)	activat*, associat*, assembl*, attach*, bound, bind*, block*, cataly*, cleav*, complex*, conjugat*, contact*, coupl*, contain*, dimer*, disassembl*, discharg*, disassociat*, down-regulat*, downregulat*, express*, inactivat*, inhibit*, induc*, interact*, mediat*, link*, modulat*, modif*, multimeri*, overexpress*, precipitat*, promot*, regulat*, repress*, stimulat*, substitut*, suppress*, transactivat*, up-regulat*, upregulat*

The asterisk (*) serves as a wildcard, matching any number of characters.

where p_1 is the number of protein pairs that contains the first category of predicates, p_2 is the number of protein pairs that contains the second category, p is the number of co-occurrence based protein pairs, i is the score corresponding to the first category of predicates, and j is the score corresponding to the second category of predicates. Compared with other text mining methods based on co-occurrence alone, this study also considered semantic information on its basis. By assigning higher weights to predicates with greater protein interactions characterization ability, the interference of meaningless information in literature can be effectively reduced, and thus more accurate protein interaction features can be screened out.

2.3. Protein attribute extraction

2.3.1. Node feature extraction

In this study, we extracted five typical attributes of proteins from experimental data and scientific literature as node features: amino acid sequences, functional descriptions, corresponding GO information, protein family information, and expression level information. For a better understanding of semantic information, we employed BERT model to learn text and sequence features of proteins and Word2Vec to learn the phrase features individually. After the pre-training of BERT, each feature was represented as a 768-dimensional vector, while Word2Vec as a 100-dimensional vector.

2.3.2. Topological structure extraction

We used several node centrality measures in the complex networks to characterize the topological structure of the PPI network, including Degree Centrality, Closeness Centrality, Betweenness and Eigenvector Centrality.

2.4. Matrix fusion

The five matrices obtained above were fused to produce a multi-feature fusion matrix, and the weights of the five features were learned automatically using an MLP (Multilayer Perceptron) process to achieve the best fusion result. The link fusion matrix M^c between protein i and protein j can be expressed as:

$$\mathrm{M}_{ij}^{c} = \alpha \mathrm{M}_{ij}^{b} + \beta \mathrm{M}_{ij}^{f} + \gamma \mathrm{M}_{ij}^{s} + \delta \mathrm{M}_{ij}^{g} + \varepsilon \mathrm{M}_{ij}^{l} \tag{2}$$

where M_{ij}^b, M_{ij}^f, M_{ij}^s, M_{ij}^g, M_{ij}^l respectively represent the biochemical, functional, structural, gene co-expression and semantic features of the protein interactions. The optimal prediction results appeared with the fusion coefficients of $\alpha = 0.047$, $\beta = 0.002$, $\gamma = 0.068$, $\delta = 0.003$, $\varepsilon = 0.082$, which were learned jointly with the GCN parameters. The attribute fusion matrix was constructed by directly jointing the five protein attribute matrices along with the topological structure feature matrix in sequence.

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2.5. Feature representation

Three feature matrices were constructed in this study: a protein node feature matrix, a multi-feature fusion protein interaction matrix, and an adjacency matrix. Among them, the attribute fusion matrix was obtained by directly splicing the six protein features extracted above in turn. The link fusion matrix was built by fusing the weights of each feature after learning them using the MLP mechanism, because this process extract feature information more effectively. The adjacency matrix contained only 0 s and 1 s, with a 0 indicating that there is no interaction between the proteins, and 1 otherwise.

2.6. Association prediction

GCN is a multilayer connected neural network architecture used to learn low-dimensional representations of nodes from graph structured data.[29] We made several adjustments to the GCN in this study in order to learn node features, edge weights, and network structure features. We introduced the three matrices that we previously obtained into the GCN encoder. Subsequently, we built a two-layer GCN to achieve multi-order neighborhood message passing, and the GCN followed the layer-wise propagation rule below:

2
2

$$H^{(l+1)} = f\left(H^{(l)}, A\right) = ReLU\left(\tilde{D}^{-\frac{1}{}} A \tilde{D}^{-\frac{1}{}} H^{(l)} W^{(l)}\right) \tag{3}$$

where $H^{(l+1)}$ is the embedding at the 1^{th}-layer, A is the adjacency matrix, $\tilde{D}$ is the degree matrix of A, $W^{(l)}$ is the weight matrix of the 1^{th}-layer. At the same time, we use ReLU as an activation function. Then we introduced sigmoid as the activation function into the GCN decoder to reconstruct the adjacency matrix for protein interactions, and the predicted association matrix can be expressed as:

$$A' = sigmoid(W \cdot E) \tag{4}$$

where $A_{i\,j}$ is the associated prediction score of protein p_i and p_j, and $W \in R^{(FF)}$ is the trainable parameter matrix. For link prediction, we evaluated the cross-entropy error over all input and output data for the adjustment of the model:

$$Loss = -\sum 1 \in_{yL} \sum_{f=1}^{F} Y_{lf} \log \sum_{i} \exp(p_i) \tag{5}$$

where y_L is the set of node indices that have links.

The GCN model used in this study has a GCN depth of 2, hidden dimensions of 128, dropout of 0.2, and a learning rate of 0.001. Some of these parameters were default parameters of the model, and some were chosen based on the model performance. Due to the fast training speed of the GCN model, the computational complexity of the modified GCN model does not vary much. Furthermore, the sparseness of the PPI network allows the method to scale to large PPI network.

The workflow of our framework was shown in Fig. 1.

3. Results

3.1. Performance evaluation

In the study, all known protein interactions were randomly divided into ten mutually exclusive subsets of the same size, and the negative samples in each subset were also randomly sampled from all the unconfirmed protein interactions with the same size. Among them, six subsets were selected as the training set, two subsets were selected as the validation set, and the remaining two were used as the test set. Link prediction was performed on the test set after training. Six metrics were used to evaluate the framework's performance: area under curve (AUC), area under precision/recall curve (AUPR), accuracy (Acc), recall (Rec), precision (Pre) and F1-score (F1). Ultimately, our predictive model reached an AUC of 0.8935, an AURP of 0.8968, an Acc of 0.7968, a Rec of 0.8334, a Pre of 0.8812 and an F1 of 0.7982.

3.2. Ablation study

To assess the contributions of various components of the framework, we conducted ablation studies comparing our model with the multi-experiment feature fusion method and the literature semantic-feature method. The results are shown in Table 2.

From these results, the multi-experiment feature fusion method and literature semantic-feature method had similar perfor- mance, with the latter performing somewhat better, indicating that both methods were

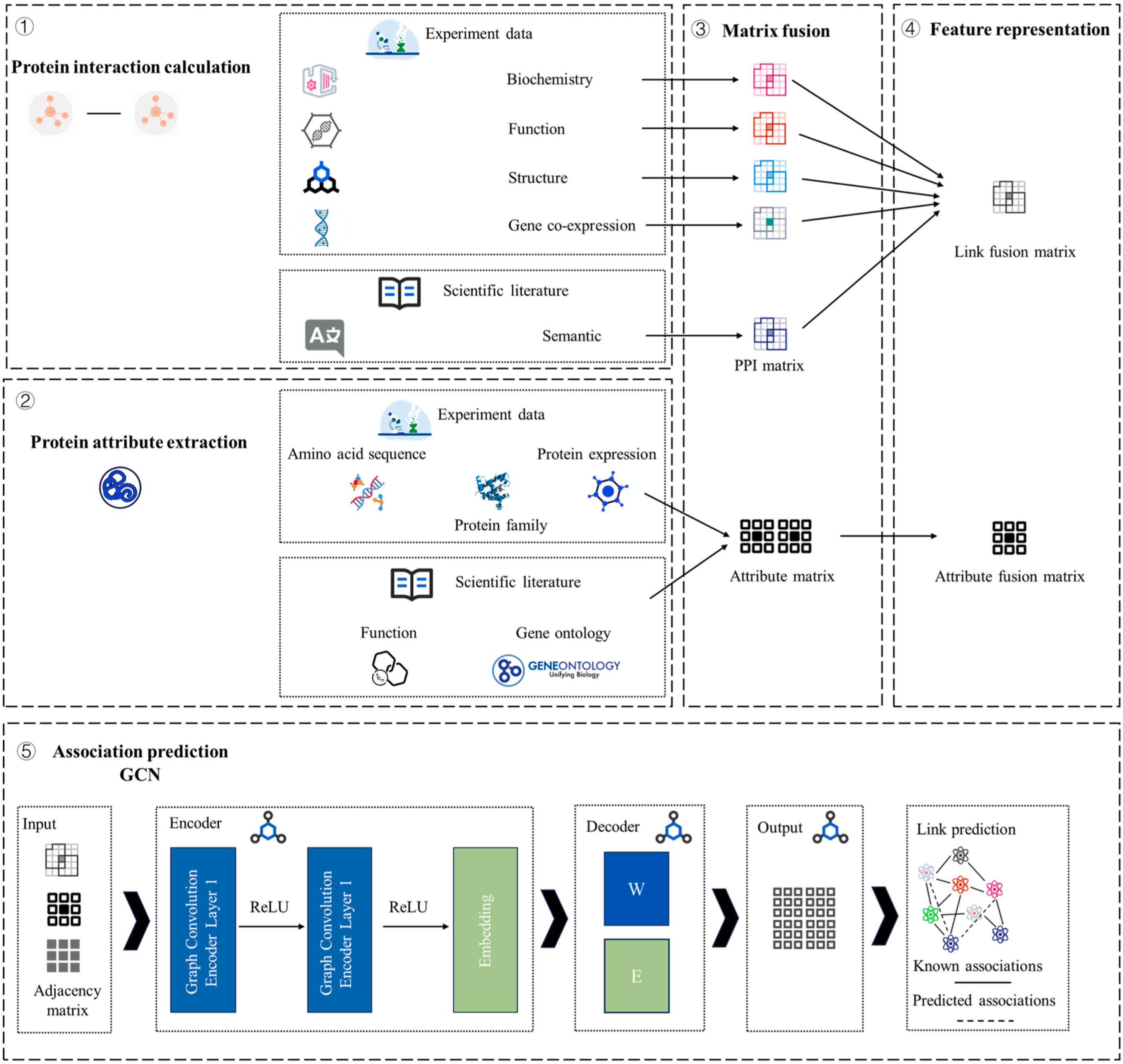

Fig. 1. A framework of our proposed method.

Table 2
Model performance comparison between multi-feature fusion and single feature. In this study, we compared multi-experiment feature fusion method and literature semantic-feature method with our proposed framework.

Methods	Feature: Biochemistry	Function	Structure	Gene co-expression	Literature semantic
Multi-experiment feature fusion method	✓	✓	✓	✓	
Literature semantic-feature method					✓
Our proposed framework	✓	✓	✓	✓	✓

Methods	AUC	AURP	Acc	Rec	Pre	F1
Multi-experiment feature fusion method	0.8343	0.8581	0.8156	0.7295	0.8836	0.6723
Literature semantic-feature method	0.8701	0.8848	0.7867	0.8217	0.7680	0.7939
Our proposed framework	0.8935	0.8968	0.7968	0.8334	0.8812	0.7982
Attributes only method	0.7580	0.7558	0.7738	0.7625	0.7176	0.7741

important for the prediction of unknown relationships. The method without considering edge weights (using attributes only) has lower values in all these metrics than the multi-experiment feature fusion method and literature semantic-feature method that incorporate both node features and edge weights. This suggests that incorporating more features of nodes and edges can help improve prediction performance of the model. The results also demonstrated that the multi-feature fusion method proposed in this study achieved the best performance with an AUC value of 0.8935, outperforming the other two methods on the majority of metrics.

3.3. Comparison with representative link prediction methods

We compared our model with four state-of-the-art link prediction methods on the same dataset. These four methods were:

- Random Walk with Restart (RWR)[30] - an algorithm that involves starting from a node in the graph and randomly choosing between returning to the start node or moving to a neighboring node at each step. Hence, this algorithm effectively captures the multifaceted relationship between two nodes, which in turn captures the overall structural information of the graph.
- DeepWalk[31] - which performs a random walk over a graph and then uses sampled sequences to learn embeddings, using the NLP based sequence models objective.
- Message Passing Neural Network(MPNN)[32] - which combines the information given by neighboring nodes based on the knowledge a node already possesses to update and obtain higher order knowledge. Generalizing a gated recurrent graph neural network provides a differentiable way to combine information from neighbors.
- GraphSAGE[33] - which learns aggregation functions for a different number of hops that are applied to sampled neighbor- hoods of different depths. These are then are used to generate node representations from the initial nodes' features.

  We also compared our model with models specifically designed for PPI prediction. The three methods were:
- KSGPPI[34] - a novel hybrid method that leverages features of protein sequences and networks to predict PPIs. The protein language model in this model were used for the extraction of more discriminative feature information from protein sequences. This model also introduced PPI network, selected the proteins in the PPI network that were most similar to the original protein sequence, and extracted their features as another feature of the original protein. This method can achieve a superior performance of PPI prediction by capturing more protein feature information and improve generalization of the model.
- BaPPI[35] - an anti-symmetric graph learning model that uses protein sequence and PPI network for protein prediction. For the utilization of protein sequence feature, this model maintains efficient and effective extraction of contextualized information. For the analysis of PPI network, this model can reduce the bias caused by the uneven distribution of node degree in PPI graph and imbalanced nature of PPI characteristics. By using the above two protein characteristics, the model demonstrates its generalizability, robustness and good performance in dealing with the imbalanced nature of PPI characteristics and the high computation cost associated with capturing long-range dependencies within protein sequence.
- GNN-PPI[36] - a graph neural network based method that uses PPI network feature for protein interaction prediction. This model proposed to incorporate correlation between proteins and excavate with a graph, in order to achieve better prediction performance. Through the introduction of correlations, the model can achieve better performance in PPI prediction, especially for the inter-novel protein interaction prediction.

According to the results in Table 3, our model outperformed the four other methods in terms of all evaluation metrics. Compared to the next-best method, GraphSAGE, our framework yielded an improvement of 2.75 % in AUC, 1.62 % in AURP, 0.94 % in Acc, 3.82 % in Rec, 9.81 % in Pre, and 0.91 % in F1 compared to the second-best model. These observations suggested that our framework has a superior ability to predict novel PPIs. Further analysis shows that the RWR method has lower Recall and Precision, indicating poor performance and difficulty in learning information in the PPI network comprehensively. The Deepwalk method has a lower Precision. The model mainly captures the graph structure in the PPI network and the learning of protein-related features is insufficient, making it difficult to achieve accurate prediction. The GraphSAGE method exhibits more balanced Precision and Recall, indicating that the model is able to comprehensively consider the protein-related features in the PPI network, resulting in a more balanced result. Our method has higher Precision than Recall. This may be due to the fact that the GCN method pays more attention to the local structure information in the PPI network, but ignores the global information. Important information in the graph is omitted, leading to node identification errors and affecting the accuracy. In addition, the framework proposed in this study, which focuses on Alzheimer's disease, is only applicable to specific diseases and lacks some generality. The reason why disease-specific contexts can enhance or limit the performance of these general link prediction baselines may be that specific diseases usually have more obvious biological patterns, such as the enrichment of certain key proteins or the formation of specific functional modules or pathways by related proteins, which make the link prediction methods more effective. However, there are significant differences in protein interaction networks for different diseases, such as data bias for rare or genetic diseases, or certain linkage prediction models rely on negative sample sampling, where unobserved interactions may be real rather than experimentally undetected, leading to lower prediction performance.

Table 3
Performance compared with existing representative link prediction methods.

Methods	AUC	AURP	Acc	Rec	Pre	F1
RWR	0.7430	0.2302	0.935	0.5291	0.3908	0.4496
DeepWalk	0.6080	0.5206	0.5965	0.7416	0.5428	0.6268
MPNN	0.8438	0.8563	0.7647	0.7737	0.7601	0.7668
GraphSAGE	0.8660	0.8806	0.7874	0.7952	0.7831	0.7891
Our proposed famework	0.8935	0.8968	0.7968	0.8334	0.8812	0.7982
KSGPPI	0.5425	0.5233	0.5517	0.4925	0.5472	5184
BaPPI	0.5364	0.3047	0.5551	0.4303	0.5140	0.5217
GNN-PP1	0.7331	0.3701	0.7628	0.5124	0.3288	0.4406
Our proposed famework	0.8935	0.8968	0.7968	0.8334	0.8812	0.7982

The general evaluation metrics of KSGPPI, BaPPI, GNN-PPI and our proposed method are shown in Table 3. We found that these models specifically designed for PPI don't work very well on our dataset. After analysis, this may be due to the fact that the above-mentioned models specifically designed for PPI prediction in the field of biomedical information usually rely on the protein sequence, structure information and the topological structure information of PPI network for prediction. By exploring the various features of the protein sequence and structure information in depth, the problems of generalizability, robustness and performance of the model can be addressed, as well as the imbalanced nature of PPI datasets. However, our model focuses more on integrating the multi-source features of PPIs to achieve prediction, considering the comprehensive performance of the model in processing various features, and focusing more on the predictive effect of protein interaction in a particular disease area. The focus of the two types of models is different. Further analysis shows that the model developed specifically for PPI is more general and applicable to human proteins of various diseases. The model can also make in-depth use of protein sequences, protein structural information and topological features of the PPI network, so as to achieve more accurate protein interaction prediction. This can also provide insights into our model. To further improve the performance of our proposed model, we can analyze the characteristics of such algorithms to improve the general applicability of the model and obtain a general framework. Then the proteins of different diseases are fine-tuned on the basis of this framework to achieve superior performance in protein interaction prediction. It is also possible to consider the integration of a wider range of features, including protein structural features, as well as the in-depth mining of protein sequences and PPI topological features for further optimization of the model.

3.4. Top ten predicted protein interactions

Turning to the actual predictions made by our framework, the top ten most probable protein interactions with respect to Alzheimer's disease are listed in Fig. 2.

4. Discussion

4.1. Superior performance

According to the aforementioned performance metrics, the method proposed in our study exhibited superior performance. This indicates that the model had a strong ability to integrate and learn from multi-dimensional information, allowing it to model system-wide PPI landscapes effectively. It is an effective strategy for potential protein interaction prediction. We found that under the modified GCN framework, the weight of biochemical, structural, and semantic features of the protein interactions are higher among the five data modality weights, indicating that these characteristics play a key role in the decision-making process of PPI prediction. These results suggest that protein structure changes and biochemical properties of proteins may be the key factors leading to abnormal interactions in the study of Alzheimer's disease. Additionally, PPI prediction models not only rely on statistics or network topology, the integration of biochemical and structural feature of protein interactions will contribute to better identification of true protein interactions. Moreover, the value also confirms that the semantic feature is an important feature in our framework and the inclusion of this feature is of great significance, which will be an enhancement of the experimental data.

4.2. Validity of the predictions

4.2.1. Analysis of model predictions

To further evaluate our results, we empirically assessed the top ten predictions. A summary of the results and their detailed explanations

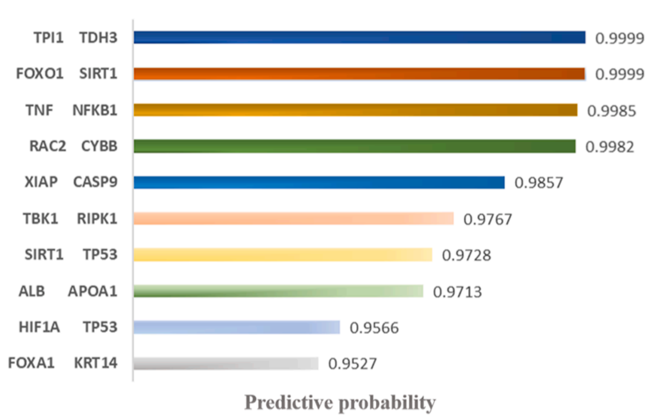

Fig. 2. The top ten predicted protein interactions with respect to Alzheimer's disease. The left side of the figure shows the pairs while the right side shows the probability of an interaction.

Table 4
Predicted list of protein interactions for Alzheimer's disease with detailed explanation.

Predictive results	Detailed explanations
FOXO1, SIRT1	Autophagy deficiency/inhibition occurs in the early stages of Alzheimer's disease. It helps to metabolize β -amyloid ($\alpha\beta$) in the Alzheimer's-affected brains. The SIRT1-FoxO pathway is an important cellular autophagy pathway. By regulating Alzheimer's-related pathways and exerting a certain protective effect, this pathway can prevent apoptosis and enhance the survival rate of neurons, making it a promising target for therapeutic interventions. However, further research is warranted.[37]
TPI1, TDH3	Recent studies suggest that using software computational tools to perform large-scale chromatographic separation mass spectrometry data have yielded robust evidence of the fact that the two proteins can indeed form a protein complex with a high level of confidence. However, the specific role of played by these protein complexes in Alzheimer's disease remains largely unexplored and the available evidence is still limited.[38]
TNF, NFKB1	Recent research has found that the formation of amyloid fibrils may be a common regulatory mechanism of signal transduction during the natural immune response. These fibrils activate the NF-kB transcriptional pathway, which can build a link between immune response and neurodegenerative diseases. This finding supports further exploration of the mechanisms underpinning Alzheimer's disease in the future.[39]
RAC2, CYBB	CYBB is a crucial protein component in the phagocytic NADPH oxidase system. Rac2 is involved in the regulation of NADPH oxidase activity and also participates in the neutrophil response in the presence of the Btk kinase. However, the exact association of this process with Alzheimer's disease remain unclear, and further exploration is needed.[40]
XIAP, CASP9	Recent research has revealed that an interplay between CASP9 and XIAP in the oocyte elimination during fetal mouse development. This process has been found to be closely related to reproductive senescence in humans. Senescence is also considered a major risk factor in the development of Alzheimer's disease because it impairs cognitive function in humans. Thus, a greater understanding of this regulatory mechanism might provide valuable insights into Alzheimer's disease.[41]
TBK1, RIPK1	Several research clues indicate that RIPK1, a member of the "receptor-interacting protein kinase" family, plays a crucial role in initiating and regulating various important physiological processes, including apoptosis, cell necrosis, and cellular inflammation. TBK1 directly binds to the cell death complex and phosphorylates RIPK1, leading to the inhibition of RIPK1 activation and the subsequent programmed cell death. Additionally, the mutations in TBK1 accelerate the aging process. Considering that Alzheimer's disease is a neurodegenerative disorder that is closely linked to the aging process, this mechanism could be closely related to Alzheimer's disease and could hold great potential as a target for future therapeutic interventions.[42]
SIRT1, TP53	Several studies demonstrate that SIRT1, as a deacetylase, has the ability to indirectly regulate post-mitotic neuronal cells by interacting with various non-histone proteins. TP53 is identified as the main target for the action of SIRT1. In certain situations such as tissue injury and DNA damage, SIRT1 can deacetylate TP53, which decreases the affinity of TP53 to bind to the regulatory sequences on DNA, known as cis-elements. Consequently, the inhibition of apoptosis allows for the preservation of neuronal cells. The investigation to this signaling cascade will hold great promise for advancing our knowledge of the pathology of Alzheimer's disease.[43]
ALB, APOA1	A study was conducted to investigate gene co-expression patterns based on the analysis of RNA-seq expression data. The results revealed that the two proteins APOA1 and ALB exhibits significant co-expression, implying a strong correlation between them. This suggests that the two proteins may share similar biological functions and involve in disease-related pathways. Specifically, the APOA1 protein is closely related to Alzheimer's disease. Therefore, the module is considered relevant to Alzheimer's disease, potentially providing valuable insights into the understanding of disease pattern.[44]
HIF1A, TP53	It has been found that direct interactions between HIF−1 alpha and Mdm2 modulate the p53 function, but the associations between Alzheimer's disease and this protein interaction have not

(continued on next page)

Table 4 (*continued*)

Predictive results	Detailed explanations
	been investi- gated within the current research scope to any great extent.[45]
FOXA1, KRT14	No study as yet has identified any interactions between these two proteins. As such, this may be a brand-new signaling pathway associated with Alzheimer's disease.

appear in Table 4.

4.2.2. Analysis on the difference between multi-feature experiment data predictions and literature data knowledge

In analyzing the similarities and differences between the predictions made based on multi-feature experiment data and literature data knowledge, we found that, generally, the probabilities predicted from multi-feature experiment data exceeded 0.9, while in the literature scored around 0.6 %. The specific percentages are given in Fig. 3. Notably, there was some overlap in the results. However, the portion was not substantial, indicating that both types of data have their own specific focuses and therefore complement each other. This validates the need to fuse the two types of data in order to achieve more meaningful knowledge discovery. As demonstrated in Table 5, eight of the top 500 predictions based on multi-feature experiment data were found in the literature. Analysis on the overlapping part revealed that (1) the predictions results based on multi-feature experiment data tended to have lower rankings in the literature and corresponded to more recent evidence source articles. This suggested that we could achieved relative accurate predictions based on multi-feature experiment data to some extent. (2) Moreover, the evidence in the literature for the predictive results based on multi-featured experiment data were clear protein interactions related to Alzheimer's disease. This indicates that the literature contained additional information beyond what the experiment data alone could provide, which could cover advanced findings in the field and serve as a complement to the experiment data.

According to Table 6, further analysis based on the difference part revealed that (1) the predictions based on multi-feature experiment data could capture potential protein interactions, which was different from the literature that were explicit record of the experiment, and this discrepancy resulted in varying outcomes. (2) Unlike the speculative results presented in the literature, where protein interactions were obtained through inference without experimental confirmation, the predictions based on multi-feature experiment data employed a computational approach to extrapolate protein interactions, and consequently the results diverged. (3) The predictions using multi-feature experiment data involved leveraging high-throughput experimental techniques to quantify protein interactions from numerous perspectives and after obtaining the precise quantitative results, inference was employed to make predictions. Whereas, the literature presented a comprehensive record of experiment findings from various angles, providing a semantic representation of protein interactions. Due to the different emphasis on what was conveyed, differences in outcomes arised. (4) It is worth noting that the literature has undergone manual review and processing, which may lead to the exclusion of certain protein interactions that have not been verified in multiple experiment or have weak correlations. Consequently, the predictions made by multi-feature experiment data might not be included in the literature, resulting in disparate results. (5) Furthermore, the literature encompassed multidirectional cross-disciplinary knowledge related to Alzheimer's disease, such as other disease information with shared mechanisms or pathways or information about other neurodegenerative disorders. It seemed challenging to learn these knowledge by inferring solely from single-dimensional experiment data, and hence contributed to the inconsistent results.

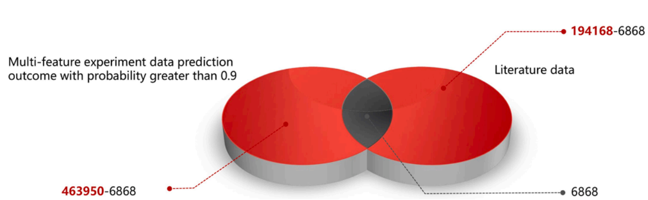

Fig. 3. The specific percentage of the similarities and differences between multi-feature experiment data predictions and literature data knowledge. Based on multi-experiment feature fusion method, we obtained 463950 predicted protein interactions with probability greater than 0.9, and in the literature data knowledge, we extracted 194168 protein interactions. The overlapping portion of the two parts was 6868.

Table 5
A list of the overlapping part of multi-feature experiment data results in top 500 predictions and literature data knowledge.

Prediction rank in experiment data	Protein interaction	Probability	Rank in literature
1	BACE1, APP	0.999	111082
Evidence in literature: BACE1 catalyzes the rate-limiting step in the generation of amyloid- β peptides. The aggregates are a proteolytic division of APP, that is, a principal component involved in the pathology of Alzheimer's disease.[46]			
1	JAK2, STAT3	0.999	141847
Evidence in literature: $\alpha\beta$ Os exert neurotoxicity and synaptotoxicity and play a critical role in the pathological progression of Alzheimer's disease. $\alpha\beta$ Oi-induced abnormalities in JAK2/STAT3 signaling pathways were reversed by hydro-xytyrosol and resulted in mildly improved cognitive function.[47]			
147	NGF, TrkB	0.997	144660
Evidence in literature: Degeneration of BFCNs in the NBM and VDB along with their connections is a key pathological event leading to memory impairment in Alzheimer's disease. Aberrant neurotrophin signaling via p75NTR contributes to BFCN dystrophy, and NGF, TrkB signaling may provide trophic support to BFCNs.[48]			
185	A2M, ApoE	0.990	113543
Evidence in literature: Levels of ApoE and A2M are significantly increased in amnestic mild cognitive impairment (aMCI). Results suggest that concentrations of certain apolipoproteins and inflammation-related factors are altered in the early stages of Alzheimer's disease.[49]			
207	BACE1, A2M	0.988	141849
Evidence in literature: The cause of Alzheimer's disease can be contributed to neuronal stress in the CNS and Abeta can be regarded as a marker. There are several stressors that generate an increase in $\alpha\beta$ expression in CNS neurons. In Amyloid Beta Synthesis, the main β -secretase in Alzheimer's disease was identified as BACE1. In $\alpha\beta$ transportation mechanisms, the ligands in this process include α2-microglobulin.[50]			
233	TLR4, CD36	0.983	134195
Evidence in literature: Curcumin reduces the imbalance of triggering receptor expressed on myeloid cells 2 and TLR4, which reduces the inflammatory response. A different study has indicated that, as Alzheimer's disease progresses, cytokines are produced in response to Aβ deposition. In conclusion, the pro- or anti- inflammatory process mediated by different cells at different stages of Alzheimer's disease may be harmful or beneficial to the organism.[51]			
338	IL−1, CXCL−10	0.972	154945
Evidence in literature: BMMSCs alleviates neuropathology and improves cognitive deficits in animal models with Alzheimer's disease. The transplantation of BMMSCs alters the local microenvironment and produces a variety of autocrine and paracrine cytokines including inflammatory cytokines(IL−1) and chemokines (CXCL−10).[52]			
460	V-CAM1, ICAM−1	0.950	165868
Evidence in literature: Blood biomarkers may be useful for the diagnosis and prediction of Alzheimer's disease brain pathology. Also, the APOE ε4 allele has shown cerebrovascular effects. Multiple genes enriched in pathways from combined blood and brain expression profiles show a significant association with the vascular injury proteins ICAM−1, SAA, and VCAM−1.[53]			

Having discussed the difference between the two types of data, it is worth noting that there was some slight overlap between the multi-feature experiment data predictions and the literature data. However, the majority of them were still different, indicating that the two data sources were somewhat associated and complementary. This could help to promote the integration of multi-directional dispersed data and contributed to a more accurate novel interaction prediction.

Table 6
Differences of the multi-feature experiment data results in top 500 predictions versus the literature data knowledge. The representative results are listed below.

Protein interactions outcomes based on multi-feature experiment data prediction	Outcome clues	Literature data
TP53, SIRT1	potential clue (PMID:35166167)	T23O, TAU
TP53, HIF1A	potential clue (PMID:10640274)	CD69, HCD2
SRC, ERBB2	potential clue (PMID:22973453)	1433 T, T23O
MAPT, GSK3B	confirmed new research (PMID:18467332, Year:2023)	HCD2, VPK6
JUN, ESR1	confirmed new research (PMID:35237381, Year:2022)	EPHB2, T23O
HDAC1, DNMT1	potential clue (PMID:17934516)	IL1B, SPRC
GSK3B, MAPT	confirmed new research (PMID:33502043, Year:2023)	HCD2, KI67
FYN, SRC	confirmed new research (PMID:34380884, Year:2022)	PSN1, TACAN
HLA-DQB1, CD74	no evidence	IL1B, PSN1
TUBB, EEF1G	no evidence	CD69, SYUA

4.3. Limitations and future work

This study has several limitations. First, the semantic information on protein interactions extracted from the literature based on co-occurrence technique and semantic predicates in this study may not be accurate enough. This is because that the predicates collected in this study were insufficient, thus leading to incomplete extraction of accurate protein interactions. Also, there is much uncertainty in scientific knowledge, which might have extracted some spurious protein interactions or even some negative protein interactions. Second, GCNs are a basic model for learning on graph-structured data. However, they are not capable of capturing all the information a graph possibly contains. Some valuable information such as rich attribute information would be overlooked, lowering the accurate of the link prediction results. Third, the PPI prediction framework in this study is mainly oriented to the field of Alzheimer's disease, without exploring its applicability to other diseases or the general PPI data, and the analysis of the model is not deep enough. However, due to the complex mechanism of Alzheimer's disease, its protein interactions involve signaling pathways, immune and inflammatory responses, protein structure, epigenetic heritance, etc., and the structural features of the protein interaction network are more complex. The protein prediction framework designed in this study can capture and learn node properties and topological structures in complex protein interaction networks. Theoretically, this framework should also perform well in other diseases and broader PPI networks, but further validation is needed. Fourth, rigorous data quality control measures were not implemented in the study to ensure the accuracy of multi-feature data. The potential biases of the literature-based data, such as publication bias, were also ignored, which would have some impact on the performance of the model. In future studies, we plan to explore a deep learning model BERT and its variant BioBERT to mine the literature more deeply for semantic information. Such models have better semantic comprehension, which allows the accurate identification of protein interactions in complex sentences. The models also have superior contextual awareness which can help to eliminate ambiguity and identify protein interactions more accurately. The models can learn rich language patterns during pre-training, allowing the processing of complex sentence structures. All of these will aid in obtaining more explicit protein interactions and reducing noise from literature-based data. We also intend to explore some other graph neural network models to try and further improve the accuracy of the link prediction task. For example, Graph Attention Network (GAT) can learn edge weights using the attention mechanism while also processing node attributes. Edge-weighted Graph Neural Network (EGNN) can directly use edge weights for graph convolution computation while processing node attributes in the meantime, inspiring future research. These specialized GNN architectures might inspire subsequent research. We will further clarify the applicability of our proposed method to general PPI data and classical benchmark datasets to provide a broader perspective on the model's utility and generalizability. Additionally, we will strictly screen and evaluate the data of various protein interaction features used in the study, especially considering the influence of literature data, to ensure the accuracy and completeness of the data, so as to further improve the accuracy of the model protein interaction prediction.

5. Conclusion

In this study, we proposed a multi-source protein features fusion framework for predicting the protein interactions associated with Alzheimer's disease. The framework fuses multi-feature of protein attributes and interactions from experiment data and literature data and constructs a weighted PPI network. An enhanced GCN that is applicable to the network then captures the various network features including node features, edge weights and topological structures and uses that knowledge to predict the most likely protein interactions. Ultimately, we got the top ten most probable novel protein interactions for Alzheimer's disease. Our results suggest that experiment data drives discovery, but the literature can complement these findings with more advanced knowledge. In summary, our framework adequately learned scattered information from both laboratories and scientific literature, which is able to provide more comprehensive and valuable information for the identification of deeper associations between proteins. This method gives researchers tremendous opportunities to understand the mechanisms of disease at the molecular level, helping them to uncover implicit knowledge with high confidence, thereby contributing to the in-depth exploration of the intricacies of Alzheimer's disease.[54] The framework designed in this study is an AI-driven method, which may bring a series of ethical issues. However, the predicted results of research are mainly used in scientific research to support scientific discovery and the proposal of new hypotheses. This problem can be avoided to some extent through standardized data processing and rational use of models. Alzheimer's disease is still a substantial public health burden nowadays. We expect our framework to be an efficient approach to reduce the time and cost of pursuing research in the laboratory and reveal confident protein interaction relationships. Protein interaction discovery helps shed light on existing treatments, potential targets, and diagnostic methods for ADthereby aiding in our understanding of the complex disease.

Supplementary information

The datasets and codes of this article are provided as supplmentary information.

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CRediT authorship contribution statement

Xiao-Li Tang: Conceptualization, Supervision. **Shi-Rui Yu:** Writing – original dralt, Methodology, Formal analysis, Conceptualization. **Xue-Mei Yang:** Writing – review & editing, Methodology. **Yi-Nan Sun:** Writing – review & editing, Proiect administration. **Yong-Jie Li:** Writing – review & editing, Project administration. **Yu-Yang Liu:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

References

1. Chen X, et al. Sequencing-based network analysis provides a core set of genes for understanding hemolymph immune response mechanisms against poly I:c stimulation in amphioctopus fangsiao. *Fish Shellfish Immunol*. 2023;133, 108544. https://doi.org/10.1016/j.fsi.2023.108544.
2. Wang S, et al. Protein-protein interaction networks as miners of biological discovery. *Proteomics*. 2022;22, e2100190. https://doi.org/10.1002/pmic.202100190.
3. Wee JJ, Kumar S. Prediction of hub genes of alzheimer's disease using a protein interaction network and functional enrichment analysis. *Genom Inf*. 2020;18, e39. https://doi.org/10.5808/GI.2020.18.4.e39.
4. Pu Y, Beck D, Verspoor K. Graph embedding-based link prediction for literature-based discovery in alzheimer's disease. *J Biomed Inform*. 2023;145, 104464. https://doi.org/10.1016/j.jbi.2023.104464.
5. Li Z, et al. Transcriptome analysis provides preliminary insights into the response of sepia esculenta to high salinity stress. *Agric Commun*. 2024;2, 100064. https://doi.org/10.1016/j.agrcom.2024.100064.
6. Nasiri E, Berahmand K, Rostami M, Dabiri M. A novel link prediction algorithm for protein-protein interaction networks by attributed graph embedding. *Comput Biol Med*. 2021;137, 104772. https://doi.org/10.1016/j.compbiomed.2021.104772.
7. Wang Y, et al. Weighted gene co-expression network analysis based on stimulation by lipopolysaccharides and polyi- nosinic:polycytidylic acid provides a core set of genes for understanding hemolymph immune response mechanisms of amphioctopus fangsiao. *Animals*. 2024;14. https://doi.org/10.3390/ani14010080.
8. Li Y, Ilie L. Predicting protein-protein interactions using sprint. *Methods Mol Biol*. 2020;2074:1–11. https://doi.org/10.1007/978-1-4939-9873-91.
9. Alekseenko A, Ignatov M, Jones G, Sabitova M, Kozakov D. *Protein–Protein and Protein–Peptide docking with ClusPro server*. US, New York, NY: Springer; 2020: 157–174.
10. Li D, et al. Princess, a protein interaction confidence evaluation system with multiple data sources. *Mol Cell Proteom*. 2008;7:1043–1052. https://doi.org/10.1074/mcp.M700287-MCP200.
11. Zdobnov EM, et al. Orthodb in 2020: evolutionary and functional annotations of orthologs. *Nucleic Acids Res*. 2021;49:D389–D393. https://doi.org/10.1093/nar/gkaa1009.
12. Hashemifar S, Neyshabur B, Khan AA, Xu J. Predicting protein-protein interactions through sequence-based deep learning. *Bioinformatics*. 2018;34:i802–i810. https://doi.org/10.1093/bioinformatics/bty573.
13. Taghipour S, Zarrineh P, Ganjtabesh M, Nowzari-Dalini A. Improving protein complex prediction by reconstructing a high-confidence protein-protein interaction network of escherichia coli from different physical interaction data sources. *BMC Bioinforma*. 2017;18:10. https://doi.org/10.1186/s12859-016-1422-x.
14. Du X, et al. Deepppi: boosting prediction of protein-protein interactions with deep neural networks. *J Chem Inf Model*. 2017;57:1499–1510. https://doi.org/10.1021/acs.jcim.7b00028.
15. Lee J, Lee D, Lee KH. Literature mining for context-specific molecular relations using multimodal representations (commodar. *BMC Bioinforma*. 2020;21:250. https://doi.org/10.1186/s12859-020-3396-y.
16. Heo GE, Xie Q, Song M, Lee J-H. Combining entity co-occurrence with specialized word embeddings to measure entity relation in alzheimer's disease. *BMC Med Inform Decis Mak*. 2019;19:240. https://doi.org/10.1186/s12911-019-0934-5.
17. Li M, Zhang H, Wang J-X, Pan Y. A new essential protein discovery method based on the integration of protein- protein interaction and gene expression data. *BMC Syst Biol*. 2012;6:15. https://doi.org/10.1186/1752-0509-6-15.
18. Zhang L, Yu G, Xia D, Wang J. Protein–protein interactions prediction based on ensemble deep neural networks. *Neurocomputing*. 2019;324:10–19. https://doi.org/10.1016/j.neucom.2018.02.097.
19. Ghasemi S, Zarei A. Improving link prediction in social networks using local and global features: a clustering-based approach. *Prog Artif Intell*. 2022;11:79–92. https://doi.org/10.1007/s13748-021-00261-3.
20. Wang C, Satuluri V, Parthasarathy S. *Local probabilistic models for link prediction. Proceedings of the 2007 Seventh IEEE International Conference on Data Mining, ICDM '07*. USA: IEEE Computer Society; 2007:322–331. https://doi.org/10.1109/ICDM.2007.108.
21. Chen B, Li F, Chen S, Hu R, Chen L. Link prediction based on non-negative matrix factorization. *PLOS ONE*. 2017;12:1–18. https://doi.org/10.1371/journal.pone.0182968.
22. Cao S, Lu W, Xu Q. *Deep neural networks for learning graph representations. Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence, AAAI'16*. AAAI Press; 2016: 1145–1152.
23. Makarov I, Kiselev D, Nikitinsky N, Subelj L. Survey on graph embeddings and their applications to machine learning problems on graphs. *PeerJ Comput Sci*. 2021;7, e357. https://doi.org/10.7717/peerj-cs.357.
24. Del Toro N, et al. The intact database: efficient access to fine-grained molecular interaction data. *Nucleic Acids Res*. 2022;50:D648–D653. https://doi.org/10.1093/nar/gkab1006.
25. Singh R, Park D, Xu J, Hosur R, Berger B. Struct2net: a web service to predict protein-protein interactions using a structure-based approach. *Nucleic Acids Res*. 2010;38:W508–W515. https://doi.org/10.1093/nar/gkq481.
26. Zhao C, Wang Z. Gogo: an improved algorithm to measure the semantic similarity between gene ontology terms. *Sci Rep*. 2018;8:15107. https://doi.org/10.1038/s41598-018-33219-y.
27. Yang S, et al. Coexpedia: exploring biomedical hypotheses via co-expressions associated with medical subject headings (mesh). *Nucleic Acids Res*. 2017;45: D389–D396. https://doi.org/10.1093/nar/gkw868.
28. Rebholz-Schuhmann D, Jimeno-Yepes A, Arregui M, Kirsch H. Measuring prediction capacity of individual verbs for the identification of protein interactions. *J Biomed Inform*. 2010;43:200–207. https://doi.org/10.1016/j.jbi.2009.09.007.
29. Zhang H, Lu G, Zhan M, Zhang B. Semi-supervised classification of graph convolutional networks with laplacian rank constraints. *Neural Process Lett*. 2022;54: 2645–2656. https://doi.org/10.1007/s11063-020-10404-7.
30. Smedley D, et al. Walking the interactome for prioritization of candidate disease genes. *Am J Hum Genet*. 2008;82:949–958. https://doi.org/10.1016/j.ajhg.2008.02.013.
31. Perozzi B, Al-Rfou R, Skiena S. *Deepwalk: Online learning of social representations. Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, KDD'14*. New York, NY, USA: Association for Computing Machinery; 2014:701–710. https://doi.org/10.1145/2623330.2623732.
32. Gilmer J, Schoenholz SS, Riley PF, Vinyals O, Dahl GE. *Neural message passing for quantum chemistry. Proceedings of the 34th International Conference on Machine Learning - Volume 70, ICML'17*. JMLR.org; 2017:1263–1272.
33. Hamilton WL, Ying R, Leskovec J. *Inductive representation learning on large graphs. Proceedings of the 31st International Conference on Neural Information Processing Systems, NIPS'17*. Red Hook, NY, USA: Curran Associates Inc.; 2017:1025–1035.
34. Hu J, Li Z, Rao B, Thafar MA, Arif M. Improving protein-protein interaction prediction using protein language model and protein network features. *Anal Biochem*. 2024;693, 115550. https://doi.org/10.1016/j.ab.2024.115550.
35. Tang T, et al. Anti-symmetric framework for balanced learning of protein–protein interactions. *Bioinformatics*. 2024;40:btae603. https://doi.org/10.1093/bioinformatics/btae603. ⟨https://academic.oup.com/bioinformatics/article-pdf/40/10/btae603/60136061/btae603.pdf⟩.
36. Lv, G., Hu, Z., Bi, Y. & Zhang, S. Learning unknown from correlations: Graph neural network for inter-novel-protein interaction prediction. *ArXiv* abs/2105.**06709** (2021).
37. Tahmi M, et al. Brain amyloid burden and resting-state functional connectivity in late middle-aged hispanics. *Front Neurol*. 2020;11, 529930. https://doi.org/10.3389/fneur.2020.529930.
38. Hu LZ, et al. Epic: software toolkit for elution profile-based inference of protein complexes. *Nat Methods*. 2019;16:737–742. https://doi.org/10.1038/s41592-019-0461-4.
39. Kleino A, et al. Peptidoglycan-sensing receptors trigger the formation of functional amyloids of the adaptor protein imd to initiate drosophila nf-b signaling. .e6 *Immunity*. 2017;47:635–647. https://doi.org/10.1016/j.immuni.2017.09.011.
40. Honda F, et al. The kinase btk negatively regulates the production of reactive oxygen species and stimulation-induced apoptosis in human neutrophils. *Nat Immunol*. 2012;13:369–378. https://doi.org/10.1038/ni.2234.
41. Liu X, Castle V, Taketo T. Interplay between caspase 9 and x-linked inhibitor of apoptosis protein (xiap) in the oocyte elimination during fetal mouse development. *Cell death amp Dis*. 2019;10:790. https://doi.org/10.1038/s41419-019-2019-x.
42. Xu D, et al. Tbk1 suppresses ripk1-driven apoptosis and inflammation during development and in aging. .e19 *Cell*. 2018;174:1477–1491. https://doi.org/10.1016/j.cell.2018.07.041.
43. Białopiotrowicz-Data E, et al. Sirt1 and hsp90 feed-forward circuit safeguards chromosome segregation integrity in diffuse large b cell lymphomas. *Cell death amp Dis*. 2023;14:667. https://doi.org/10.1038/s41419-023-06186-0.
44. van Dam S, Võsa U, van der Graaf A, Franke L, de Magalhães JP. Gene co-expression analysis for functional classification and gene-disease predictions. *Brief Bioinforma*. 2018;19:575–592. https://doi.org/10.1093/bib/bbw139.
45. Chen D, Li M, Luo J, Gu W. Direct interactions between hif-1 alpha and mdm2 modulate p53 function. *J Biol Chem*. 2003;278:13595–13598. https://doi.org/10.1074/jbc.c200694200.
46. Dey J, et al. Electrochemical detection of alzheimer's disease biomarker, -secretase enzyme (bace1), with one-step synthesized reduced graphene oxide. *Front Bioeng Biotechnol*. 2022;10, 873811. https://doi.org/10.3389/fbioe.2022.873811.
47. Araki W, Kametani F. Protection against amyloid- oligomer neurotoxicity by small molecules with antioxidative properties: potential for the prevention of Alzheimer's disease dementia. *Antioxidants*. 2022;11:132. https://doi.org/10.3390/antiox11010132.
48. Gonzalez S, et al. Small molecule modulation of trkb and trkc neurotrophin receptors prevents cholinergic neuron atrophy in an Alzheimer's disease mouse model at an advanced pathological stage. *Neurobiol Dis*. 2022;162, 105563. https://doi.org/10.1016/j.nbd.2021.105563.
49. Wang T, et al. Association of plasma apolipoproteins and levels of inflammation-related factors with different stages of Alzheimer's disease: a cross-sectional study. *BMJ Open*. 2022;12, e054347. https://doi.org/10.1136/bmjopen-2021-054347.
50. Decourt B, et al. The cause of alzheimer's disease: the theory of multipathology convergence to chronic neuronal stress. *Aging Dis*. 2022;13:37–60. https://doi.org/10.14336/ad.2021.0529.
51. Wan T, et al. Research progress on mechanism of neuroprotective roles of apelin-13 in prevention and treatment of Alzheimer's disease. *Neurochem Res*. 2022;47: 205–217. https://doi.org/10.1007/s11064-021-03448-1.
52. Qin C, Bai L, Li Y, Wang K. The functional mechanism of bone marrow-derived mesenchymal stem cells in the treatment of animal models with alzheimer's disease:

crosstalk between autophagy and apoptosis. *Stem Cell Res amp Ther*. 2022;13:90. https://doi.org/10.1186/s13287-022-02765-8.
53. Panitch R, et al. Blood and brain transcriptome analysis reveals apoe genotype-mediated and immune-related pathways involved in Alzheimer disease. *Alzheimer'S Res amp Ther*. 2022;14:30. https://doi.org/10.1186/s13195-022-00975-z.
54. Khazaal SM, Maarouf H. Predicting coronary artery disease utilizing support vector machines: optimizing predictive model. *Mesop J Artif Intell Health*. 2023;2023: 21–26. https://doi.org/10.58496/MJAIH/2023/004.